

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas <i>Master of Science in Computer Engineering</i>	Aug 2024 – May 2026 <i>Fayetteville, AR</i>
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques	
Sri Ramakrishna Engineering College <i>Bachelor of Engineering in Electronics and Communication</i>	Aug 2019 – May 2023 <i>India</i>

TECHNICAL SKILLS

Cloud and Tools: Component architecture and design systems, Headless CMS (e.g. Contentful), Prompt engineering, Structured data and technical SEO, AI-assisted development workflows, Git, Docker, Linux, HTML, CSS, CI/CD, JavaScript

Machine Learning and AI: LangChain, RAG, Flask, Streamlit

Data Engineering:

Languages: SQL, Python

PROFESSIONAL EXPERIENCE

Data Engineering Intern <i>InsOps Inc.</i>	Jul 2024 – Present <i>Chicago, Illinois</i>
- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics - Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics	
Teaching Assistant <i>University of Arkansas, Dept. of Computer Science and Engineering</i>	Sep 2024 – May 2026 <i>Fayetteville, AR</i>
Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals	
Research Assistant <i>AICV Lab, University of Arkansas</i>	Sep 2024 – Jan 2026 <i>Fayetteville, AR</i>
- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation - Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID	
Project Engineer <i>Soliton Technologies</i>	Jun 2023 – Jun 2024 <i>India</i>
- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation - Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements	

PROJECTS

Local LLM-Powered News Research Platform	<i>Python, LangChain, Streamlit, FAISS, Ollama, Llama 3.2</i>
- Built an end-to-end local ETL pipeline using Streamlit and WebBaseLoader to ingest, clean, chunk, and preprocess unstructured web content on demand - Designed a local vector store with FAISS and nomic-embed-text to serialize high-dimensional embeddings, bypassing redundant ingestion and reducing processing latency - Implemented a RAG pipeline via LangChain orchestrating semantic queries against vectorized data using local Llama 3.2 instances	
Resume Classification Model	<i>Python, Scikit-learn, Pandas, TF-IDF, Logistic Regression, Flask</i>
- Built an end-to-end NLP classification pipeline using TF-IDF feature extraction and Logistic Regression across 25 job-role categories from 962 unstructured resumes - Achieved 89.48% test accuracy and 88% macro F1 across 193 held-out test samples - Deployed the model through a Flask REST API for real-time prediction	